

IsoDecipher — APA Trajectory Analysis Framework

Concept

Along pseudotime, cells change their polyA site usage (PUI). This reflects dynamic 3' UTR regulation during differentiation — a signal invisible to standard scRNA-seq pipelines.

Known biology:

- T/B cell activation → 3' UTR shortening (PUI↑, proximal site preferred)
- Terminal differentiation → 3' UTR lengthening (PUI↓, distal site preferred)
- Plasma cell secretion switching → IGHM_G0↑ (secreted isoform dominates)

No existing tool does this analysis. IsoDecipher is uniquely positioned to enable it.

Algorithm Design

Step 1: Compute PUI per cell per gene

```
import numpy as np
import pandas as pd
import scanpy as sc

def compute_pui(adata, gene, g0_label='_G0', g1_label='_G1', pseudocount=1e-10):
    """
    Compute Proximal Usage Index (PUI) per cell for a given gene.
    PUI = G0_counts / (G0_counts + G1_counts)
    PUI → 1: proximal site dominant (short 3' UTR)
    PUI → 0: distal site dominant (long 3' UTR)
    """
    g0 = gene + g0_label
    g1 = gene + g1_label

    counts_g0 = adata[:, g0].X.toarray().flatten()
    counts_g1 = adata[:, g1].X.toarray().flatten()

    pui = counts_g0 / (counts_g0 + counts_g1 + pseudocount)
    return pui
```

```
# Add PUI for all genes in panel
gene_list = ['IGHM', 'CD274', 'CD44', 'PDCD1'] # from IsoDecipher panel
for gene in gene_list:
    adata.obs[f'{gene}_PUI'] = compute_pui(adata, gene)
```

Step 2: Get pseudotime axis

```
# Option A: scVelo latent time (preferred for dynamic data)
import scvelo as scv
scv.tl.latent_time(adata)
adata.obs['pseudotime'] = adata.obs['latent_time']

# Option B: Diffusion Pseudotime (DPT)
sc.tl.dpt(adata)
adata.obs['pseudotime'] = adata.obs['dpt_pseudotime']
```

Step 3: Bin cells by pseudotime

```
def bin_by_pseudotime(adata, n_bins=20):
    """Divide cells into pseudotime bins."""
    adata.obs['pt_bin'] = pd.cut(
        adata.obs['pseudotime'],
        bins=n_bins,
        labels=False
    )
    return adata

adata = bin_by_pseudotime(adata, n_bins=20)
```

Step 4: Compute mean PUI per bin

```
def compute_pui_trajectory(adata, gene_list):
    """
    Compute mean PUI per pseudotime bin for each gene.
    Returns DataFrame: bins × genes
    """
    results = {}
    for gene in gene_list:
```

```

    pui_col = f'{gene}_PUI'
    if pui_col in adata.obs.columns:
        traj = adata.obs.groupby('pt_bin')[pui_col].mean()
        results[gene] = traj
    return pd.DataFrame(results)

pui_traj = compute_pui_trajectory(adata, gene_list)

```

Step 5: Smooth curve fitting

```

from scipy.signal import savgol_filter

def smooth_trajectory(traj_df, window=5, polyorder=2):
    """Apply Savitzky-Golay smoothing to PUI trajectories."""
    smoothed = traj_df.apply(
        lambda col: savgol_filter(col.fillna(col.mean()), window, polyorder)
    )
    return pd.DataFrame(smoothed, index=traj_df.index, columns=traj_df.columns)

pui_traj_smooth = smooth_trajectory(pui_traj)

```

Step 6: Statistical test — which genes have significant APA dynamics?

```

from scipy.stats import spearmanr

def test_apa_dynamics(adata, gene_list, min_cells=10):
    """
    Test which genes show significant PUI correlation with pseudotime.
    Returns DataFrame with Spearman rho and p-value per gene.
    """
    results = []
    pt = adata.obs['pseudotime'].values

    for gene in gene_list:
        pui_col = f'{gene}_PUI'
        if pui_col not in adata.obs.columns:
            continue
        pui = adata.obs[pui_col].values

        # Remove NaN
        mask = ~np.isnan(pui) & ~np.isnan(pt)

```

```

    if mask.sum() < min_cells:
        continue

    rho, pval = spearmanr(pt[mask], pui[mask])
    results.append({
        'gene': gene,
        'spearman_rho': rho,
        'pvalue': pval,
        'direction': 'shortening' if rho > 0 else 'lengthening'
    })

df = pd.DataFrame(results)
df['significant'] = df['pvalue'] < 0.05
return df.sort_values('pvalue')

apa_stats = test_apa_dynamics(adata, gene_list)
print(apa_stats)

```

Step 7: Visualization

```

import matplotlib.pyplot as plt
import seaborn as sns

def plot_apa_trajectory(pui_traj_smooth, genes=None, figsize=(12, 4)):
    """
    Plot PUI trajectory along pseudotime for selected genes.
    """
    if genes is None:
        genes = pui_traj_smooth.columns.tolist()

    fig, axes = plt.subplots(1, len(genes), figsize=figsize)
    if len(genes) == 1:
        axes = [axes]

    for ax, gene in zip(axes, genes):
        ax.plot(pui_traj_smooth.index, pui_traj_smooth[gene],
                linewidth=2, color='royalblue')
        ax.fill_between(pui_traj_smooth.index, pui_traj_smooth[gene],
                        0.5, alpha=0.2, color='royalblue')
        ax.axhline(0.5, linestyle='--', color='gray', alpha=0.5)
        ax.set_xlabel('Pseudotime')
        ax.set_ylabel('PUI (Proximal Usage Index)')
        ax.set_title(f'{gene} APA Trajectory')
        ax.set_ylim(0, 1)

```

```

plt.tight_layout()
return fig

def plot_apa_heatmap(pui_traj_smooth, figsize=(12, 6)):
    """
    Heatmap of PUI dynamics across pseudotime for all genes.
    Rows = genes, Columns = pseudotime bins
    """
    fig, ax = plt.subplots(figsize=figsize)
    sns.heatmap(
        pui_traj_smooth.T,
        cmap='RdBu_r',
        center=0.5,
        vmin=0, vmax=1,
        ax=ax,
        cbar_kws={'label': 'PUI'})
    ax.set_xlabel('Pseudotime bins')
    ax.set_ylabel('Gene')
    ax.set_title('APA Dynamics Along Pseudotime')
    plt.tight_layout()
    return fig

# Plot
fig1 = plot_apa_trajectory(pui_traj_smooth, genes=['IGHM', 'CD274', 'CD44'])
fig2 = plot_apa_heatmap(pui_traj_smooth)

```

Output Summary

Output	Description
<code>{gene}_PUI</code> in <code>adata.obs</code>	Per-cell PUI for each gene
<code>pui_traj</code> DataFrame	Mean PUI per pseudotime bin
<code>pui_traj_smooth</code> DataFrame	Smoothed PUI trajectory
<code>apa_stats</code> DataFrame	Spearman rho + p-value per gene
Trajectory plots	PUI vs pseudotime per gene
Heatmap	All genes × pseudotime bins

Extension Ideas

1. **Waddington-style UMAP** — rotate UMAP so pseudotime aligns with y-axis (naive top, differentiated bottom)
 2. **APA × RNA velocity** — overlay PUI dynamics with velocity streamlines
 3. **Condition comparison** — compare APA trajectories between stimulated vs unstimulated cells
 4. **Checkpoint gene focus** — PD-L1 (CD274) APA dynamics in tumor microenvironment
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Why This Is Novel

- No existing tool connects pseudotime with APA dynamics at single-cell resolution
- Direct extension of IsoDecipher — same count matrix, new biological layer
- Directly applicable to your B cell → plasma cell differentiation data
- Preprint-worthy: “Dynamic APA regulation during immune cell differentiation revealed by IsoDecipher”